

4.25

a.

The intercept of 4.39(thousands of dollars) implies a base value for a 0-square-foot house, which has no practical economic meaning. The coefficient for SQFT is 0.036044 , which represents a semi-elasticity. This means that for every additional 1 square foot of area, the expected price of the house increases by approximately 0.036%.

- Slope at sample mean: 9.019661
- Elasticity at sample mean 0.9833701

b.

The intercept of 2.04965 estimates the log-price of a 1-square-foot house, which has no meaningful economic interpretation. The coefficient for $\ln(\text{SQFT})$ is 1.02483, which represents the constant elasticity of price with respect to square footage. This means that for every 1% increase in house size, the expected price of the house increases by 1.02%.

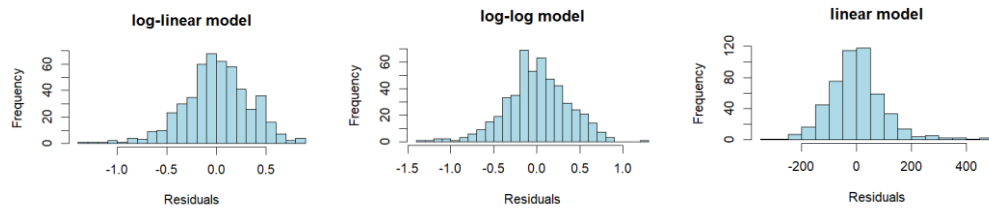
- Slope at sample mean: 9.399923
- Elasticity at sample mean 1.024828

c.

- Linear Model R-squared: 0.6413167
- Generalized R-squared for Log-Linear (a): 0.6621612
- Generalized R-squared for Log-Log (b): 0.6445084

The standard R^2 for the linear model is 0.6413167. After transforming the predictions back to dollar terms, the generalized R^2 for the log-linear model (a) is 0.6621612, and the generalized R^2 for the log-log model (b) is 0.6445084. Comparing these three values, the log-linear model has the highest value. This indicates that it provides the best overall fit and explains the highest proportion of the variation in actual house prices.

d.



Jarque-Bera Test for the residuals of the log-linear model

- X-squared = 26.679 | df = 2 | p-value = 1.61e-06
- Reject the null hypothesis. The distribution of the residuals is not compatible with the assumption of normality.

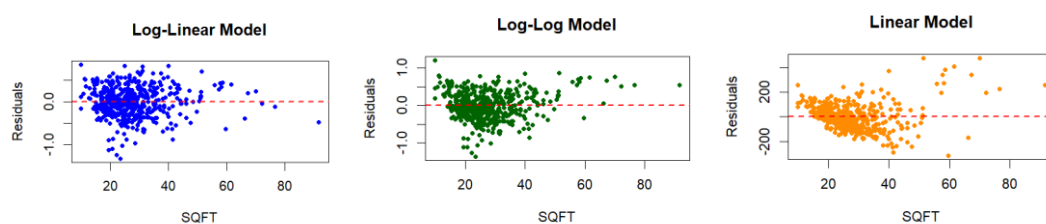
Jarque-Bera Test for the residuals of the log-log model

- X-squared = 14.279 | df = 2 | p-value = 0.0007933
- Reject the null hypothesis. The distribution of the residuals is not compatible with the assumption of normality.

Jarque-Bera Test for the residuals of the linear model

- X-squared = 221.11 | df = 2 | p-value < 2.2e-16
- Reject the null hypothesis. The distribution of the residuals is not compatible with the assumption of normality.

e.



None of the models displays a perfectly random, homoskedastic scatter of residuals. The linear model suffers from increasing variance as house size grows, while the log transformations appear to have overcorrected, resulting in higher variance for smaller homes.

f.

- Model (a) Log-Linear (Naive): 214.2336 (thousands of dollars)
- Model (a) Log-Linear (Corrected): 226.9817 (thousands of dollars)

- Model (b) Log-Log (Naive): 227.5386 (thousands of dollars)
- Model (b) Log-Log (Corrected): 243.1514 (thousands of dollars)

- Model (c) Linear: 246.4557 (thousands of dollars)

g.

- Model (a) Log-Linear Model:
 - 95% PI: [109.7685, 418.1167].
- Model (b) Log-Log Model:
 - 95% PI: [111.1406, 465.8406].
- Model (c) Linear Model:
 - 95% PI: [44.27727, 448.6341].

h.

Based on the diagnostic results above, none of the models provide a perfect fit. As indicated by the Jarque-Bera tests, the residuals of all three models reject the null hypothesis of normality. However, compared to other models, the Log-Linear model is the better one. The rationale is as follows:

- The Log-Linear model yields the highest generalized R^2 . This indicates that it captures the highest proportion of the variance in actual house prices.
- Compared to the model b and model c, the Log-Linear model significantly compresses the variance caused by extreme outliers.
- The logarithmic transformation ensures that both the predicted house prices and the lower bounds of the prediction intervals remain strictly positive.

Therefore, despite the violation of the normality assumption, the Log-Linear model remains the most robust and practically useful choice for this dataset, especially given the large sample size."